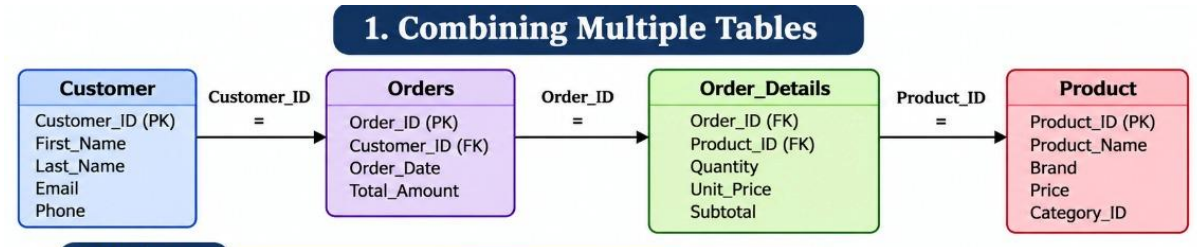
DATABASE RELATIONSHIP ANALYSIS USING JOINS

Week: 8

1.Combining Multiple Tables

The EcoCart database contains different tables that store customer, product, order, and payment information.

The tables can be combined using JOIN operations to retrieve related information



SQL Query

```
SELECT C.Customer_ID, C.First_Name, C.Last_Name,
       O.Order_ID, P.Product_ID, P.Product_Name,
       OD.Quantity, OD.Unit_Price, OD.Subtotal
FROM Customer C
JOIN Orders O ON C.Customer_ID = O.Customer_ID
JOIN Order_Details OD ON O.Order_ID = OD.Order_ID
JOIN Product P ON OD.Product_ID = P.Product_ID
ORDER BY O.Order_ID;
```

Explanation

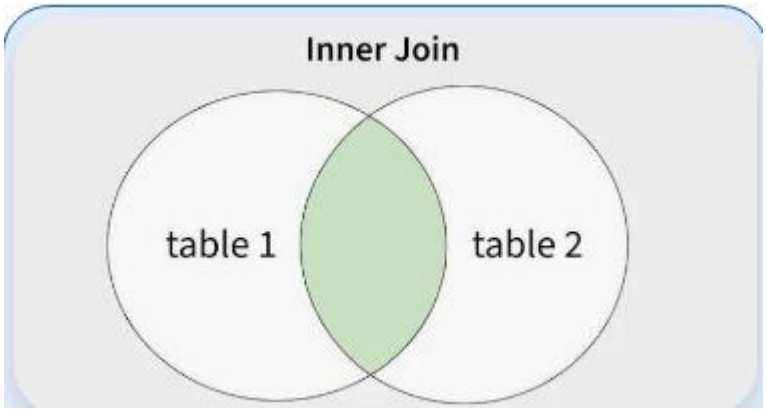
This query combines four tables: **Customer, Orders, Order_Details** and **Product**. It displays customer details, order details and product information.

Sample Output

Customer_ID	First_Name	Last_Name	Order_ID	Product_ID	Product_Name	Quantity	Unit_Price	Subtotal
101	Keerthana	Parthiban	501	1001	Laptop	1	45000.00	45000.00
101	Keerthana	Parthiban	501	1002	Mouse	2	500.00	1000.00
102	Divya	Sharma	502	1003	Mobile	1	20000.00	20000.00

2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date,
  6         O.Total_Amount
  7  FROM Customer C
  8  JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID
 10  ORDER BY O.Order_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
         201 Arun
Kumar                                                 501 06-OCT-26
       84999

         202 Priya
Sharma                                                502 06-OCT-26
       64999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

         203 Rahul
```

2.2 Orders and Payment

```
SQL> SELECT O.Order_ID,
  2         O.Customer_ID,
  3         O.Order_Date,
  4         O.Total_Amount,
  5         P.Payment_ID,
  6         P.Payment_Method,
  7         P.Payment_Status,
  8         P.Amount
  9  FROM Orders O
 10  JOIN Payment P
 11  ON O.Order_ID = P.Order_ID
 12  ORDER BY O.Order_ID;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID
---------- ----------- --------- ------------ ----------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT
------------------------------ -------------------- ----------
       501         201 06-OCT-26        84999        701
UPI                            Successful                84999

       502         202 06-OCT-26        64999        702
Credit Card                    Successful                  799

       503         203 06-OCT-26        45999        703
Debit Card                     Failed                    45999
```

2.3 Product and Order_Details

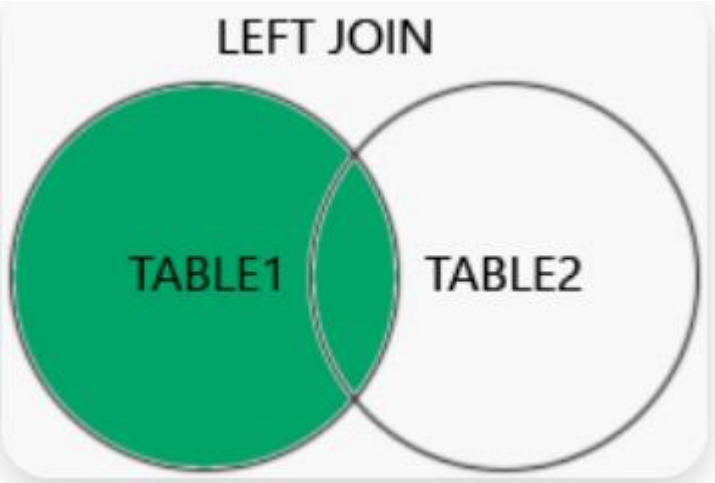
```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         P.Brand,
  4         P.Price,
  5         OD.Order_Detail_ID,
  6         OD.Order_ID,
  7         OD.Quantity,
  8         OD.Unit_Price,
  9         OD.Subtotal
 10  FROM Product P
 11  JOIN Order_Details OD
 12  ON P.Product_ID = OD.Product_ID
 13  ORDER BY OD.Order_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE ORDER_DETAIL_ID   ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- --------------- ---------- ---------- ---------- ----------
       101
iPhone 15
Apple
     74999            1001        501          1      79999      79999


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

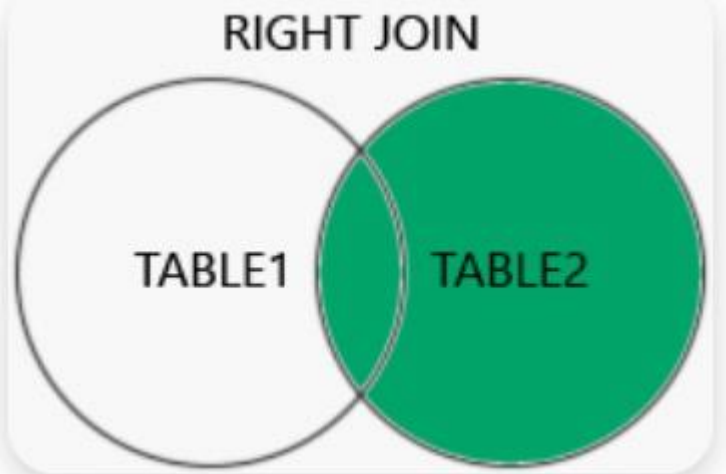
```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         C.Email,
  5         O.Order_ID,
  6         O.Order_Date,
  7         O.Total_Amount
  8  FROM Customer C
  9  LEFT JOIN Orders O
 10  ON C.Customer_ID = O.Customer_ID
 11  ORDER BY C.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
       201 Arun
Kumar
arun@gmail.com
       501 06-OCT-26        84999


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT O.Order_ID,
  2         O.Customer_ID,
  3         C.First_Name,
  4         C.Last_Name,
  5         C.Email,
  6         O.Order_Date,
  7         O.Total_Amount
  8  FROM Orders O
  9  LEFT JOIN Customer C
 10  ON O.Customer_ID = C.Customer_ID
 11  ORDER BY O.Order_ID;

  ORDER_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
       501         201 Arun
Kumar
arun@gmail.com
06-OCT-26        84999


  ORDER_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
ORDER_DAT TOTAL_AMOUNT
```

5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         C.Customer_ID,
  4         C.First_Name,
  5         C.Last_Name,
  6         P.Product_ID,
  7         P.Product_Name,
  8         OD.Quantity,
  9         OD.Unit_Price,
 10         OD.Subtotal,
 11         O.Total_Amount
 12  FROM Orders O
 13  JOIN Customer C
 14  ON O.Customer_ID = C.Customer_ID
 15  JOIN Order_Details OD
 16  ON O.Order_ID = OD.Order_ID
 17  JOIN Product P
 18  ON OD.Product_ID = P.Product_ID
 19  ORDER BY O.Order_ID;

  ORDER_ID ORDER_DAT CUSTOMER_ID
---------- --------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT
---------- ---------- ---------- ------------
       501 06-OCT-26         201
Arun
```

6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date,
  6         P.Product_Name,
  7         OD.Quantity,
  8         OD.Unit_Price,
  9         OD.Subtotal
 10  FROM Customer C
 11  JOIN Orders O
 12  ON C.Customer_ID = O.Customer_ID
 13  JOIN Order_Details OD
 14  ON O.Order_ID = OD.Order_ID
 15  JOIN Product P
 16  ON OD.Product_ID = P.Product_ID
 17  ORDER BY C.Customer_ID, O.Order_Date;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
         201 Arun
Kumar                                                  501 06-OCT-26
iPhone 15
         1      79999      79999


CUSTOMER_ID FIRST_NAME
```

7. Customer Purchase History with Payment

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date,
  6         P.Product_Name,
  7         OD.Quantity,
  8         OD.Subtotal,
  9         PY.Payment_Method,
 10         PY.Payment_Status,
 11         PY.Amount
 12  FROM Customer C
 13  JOIN Orders O
 14  ON C.Customer_ID = O.Customer_ID
 15  JOIN Order_Details OD
 16  ON O.Order_ID = OD.Order_ID
 17  JOIN Product P
 18  ON OD.Product_ID = P.Product_ID
 19  JOIN Payment PY
 20  ON O.Order_ID = PY.Order_ID
 21  ORDER BY C.Customer_ID, O.Order_Date;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- ------------------------------ --------------------
    AMOUNT
----------
```

8. Multi-Table Business Report

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date,
  6         P.Product_Name,
  7         P.Brand,
  8         OD.Quantity,
  9         OD.Unit_Price,
 10         OD.Subtotal,
 11         PY.Payment_Method,
 12         PY.Payment_Status,
 13         PY.Amount
 14  FROM Customer C
 15  JOIN Orders O
 16  ON C.Customer_ID = O.Customer_ID
 17  JOIN Order_Details OD
 18  ON O.Order_ID = OD.Order_ID
 19  JOIN Product P
 20  ON OD.Product_ID = P.Product_ID
 21  JOIN Payment PY
 22  ON O.Order_ID = PY.Order_ID
 23  ORDER BY O.Order_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_METHOD
```

9. Payment Transaction Report

```
SQL> SELECT P.Payment_ID,
  2         P.Order_ID,
  3         C.Customer_ID,
  4         C.First_Name,
  5         C.Last_Name,
  6         P.Payment_Method,
  7         P.Payment_Date,
  8         P.Payment_Status,
  9         P.Amount
 10  FROM Payment P
 11  JOIN Orders O
 12  ON P.Order_ID = O.Order_ID
 13  JOIN Customer C
 14  ON O.Customer_ID = C.Customer_ID
 15  ORDER BY P.Payment_ID;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_D PAYMENT_STATUS           AMOUNT
------------------------------ --------- -------------------- ----------
       701        501         201
Arun
Kumar
UPI                            04-NOV-26 Successful                84999


PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
```